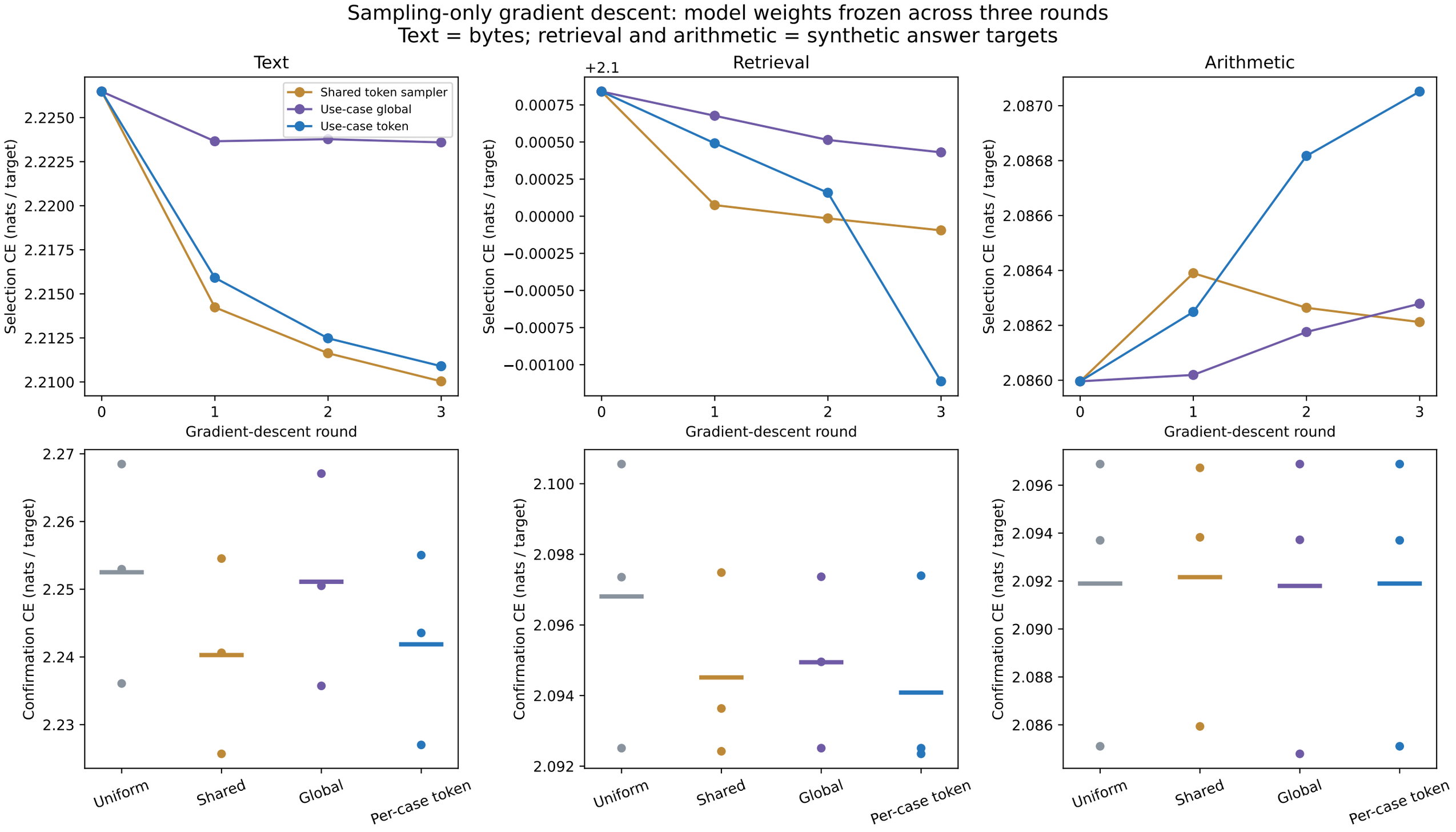
Sampling-only gradient descent: model weights frozen across three rounds
Text = bytes; retrieval and arithmetic = synthetic answer targets
Text
Retrieval
Arithmetic
Shared token sampler
Use-case global
Use-case token
Selection CE (nats / target)
Gradient-descent round
+2.1
Confirmation CE (nats / target)
Uniform
Shared
Global
Per-case token